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Review

Lower extremity return to sport testing: A systematic review

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ABSTRACT

Background: Lower extremity injuries account for an enormous portion of sports medicine cases in the United States each year. Unfortunately, there are no uniform criteria for athletes to complete prior to returning to sport (RTS) following a lower extremity injury. Therefore, the purpose of this systematic review is to review current literature for joint-specific and global lower extremity testing to determine the most valid functional test that can be utilized to reduce the risk of re-injury as athletes RTS.
Methods: A systematic search of PubMed, PubMed Central, Cochrane Library, OVID, and Embase databases was conducted for studies prior to May 2024 following PRISMA guidelines. ROBINS-I Tool was utilized for the risk of bias assessment.
Results: Of 19,189 studies, 114 (0.6%) studies published prior to May of 2024 met inclusion criteria and were analyzed. Eighty five percent of articles discussed RTS for individuals with knee pathology. Furthermore, 82% specifically analyzed RTS following ACL reconstruction. The most common RTS test was isokinetic dynamometry testing which is seen in 73% of studies. Only 6.2% of studies analyzed RTS for individuals with hip pathology and only two studies analyzed RTS for patients with ankle injuries.
Conclusion: Even with the enormous amount of literature that exists regarding ACL injuries and testing there is no standardized criterion for RTS clearance. The suggested test batteries from this review can serve as a framework for future research and validation for joint-specific RTS functional testing.
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1. Introduction

Lower extremity injuries account for an enormous portion of sports medicine cases. Anterior cruciate ligament (ACL) injuries alone account for greater than 50% of all knee injuries affecting over 200,000 people per year in the United States [1–3]. Unfortunately, even after a lengthy recovery process, young athletes (<20–25 year) who experience ACL tears have a higher incidence of reinjury when returning to high-risk sport (23%) [4]. At least a portion of the high re-injury rate can be attributed to the lack of a standardized return to sport (RTS) protocols [5].

This absence of RTS testing criteria is further complicated by the lack of a standardized definition of the term "return to sport." Most commonly "return to sport" is defined as playing in at least one game of the athletes' respective sport, or the first game played after treatment [6]. The timeline to RTS and medical clearance varies amongst each physician or institution and is also dependent upon injury, treatment, and athlete-specific characteristics. Previous research has attempted uncover and unify lower extremity RTS testing, especially following ACL reconstruction (ACLR) where the gold standard for strength testing is still thought to be isokinetic dynamometry [7–10]. Additional tests that have been studied include single leg hop for distance (SLHD), triple hop for distance (THD), cross-over hop for distance (CHD), and 6-meter (6-m) timed hop [11]. Hurley et al. assessed variations of these tests for ACLR RTS with a systematic review incorporating 8 studies and reported that passing RTS testing following ACLR will lower ACL graft rupture but not contralateral ACL injury [12]. This study highlights the importance of graft consideration as well as athlete function, strength, and confidence, but does not describe or suggest a standardized RTS testing battery based on their results. Aside from knee RTS testing, there is a paucity of functional testing research for hip and ankle pathologies. Therefore, the purpose of this systematic review is to review current literature for joint-specific and global lower extremity testing to determine the most valid functional test or series of tests that can be utilized to prevent re-injury as athletes RTS. It is hypothesized that there is not a uniform RTS criteria and more research is required to identify the most valid functional test batteries for joint-specific RTS lower extremity testing.

2. Methods

2.1. Search strategy and study selection

This study was conducted in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Institutional review board approval was not required as this study was a systematic review of published studies. A manual study selection was performed using the PubMed, PubMed Central, Cochrane Library, OVID, and Embase. The following search terms were utilized: "return to sport" OR "RTS" OR "functional test" OR "landing task" OR "landing mechanics" OR hop test" AND "sport" AND "hip" OR "knee" OR "ankle" OR "lower extremity". These databases were searched from inception to October 2021. A second search from October 2021 to May 2024 was conducted to include newly published literature.

2.2. Eligibility criteria

Studies that met the inclusion criteria were published in the English language and used non-cadaveric human subjects. Both operative and non-operative treatments were included in this review. Knee injuries that were used to meet inclusion criteria included: ACL, MPFL, meniscus repair, multi-ligament knee reconstructions, and cartilage restoration. Hip injuries

Table 1
Risk of Bias Assessment via ROBINS-I Tool.

Risk of Bias Assessment of Included Studies using the ROBINS-I Tool [13]								
Authors	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviation from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	Overall Bias
Aizawa J, et al. [55] (2022)	moderate	low	low	low	low	low	low	low
Aizawa J, et al. [56] (2024)	moderate	moderate	low	low	low	low	low	low
Albano, T et al. [38] (2020)	severe	moderate	moderate	low	low	low	low	severe
Alswat, M et al. [57] (2020)	severe	low	moderate	low	low	low	moderate	severe
Alzakerin, H et al. [58] (2020)	severe	moderate	moderate	low	low	low	low	severe
Aujla, R.S. et al. [59] (2021)	moderate	severe	moderate	low	low	moderate	low	moderate
Birchmeier, T et al. [25] (2019)	severe	moderate	moderate	low	low	low	low	severe
Bodkin, S et al. [60] (2017)	severe	moderate	moderate	low	low	moderate	low	severe
Bodkin, S et al. [34] (2018)	moderate	moderate	moderate	low	low	low	low	moderate
Borawski J, et al. [61] (2024)	moderate	moderate	low	low	low	moderate	moderate	moderate
Browning, R et al. [51] (2021)	moderate	moderate	low	low	moderate	low	moderate	moderate
Brunst, C et al. [62] (2020)	moderate	low	moderate	low	moderate	low	moderate	moderate
Burland, J et al. [39] (2018)	moderate	moderate	low	low	low	moderate	low	moderate
Burnham, J et al. [63] (2016)	severe	severe	low	low	moderate	low	low	severe
Byrne, L et al. [64] (2021)	moderate	moderate	low	low	low	low	low	moderate
Byrne, L et al. [64] (2021)	low	low	low	low	moderate	moderate	moderate	low
Caffrey, E et al. [53] (2009)	moderate	moderate	low	low	moderate	low	low	moderate
Cristiani, R et al. [65] (2019)	severe	moderate	severe	low	low	low	moderate	severe
Cronstrom, A et al. [66] (2023)	moderate	severe	low	low	moderate	moderate	low	moderate
Crotty, N.M.N. et al. [67] (2022)	moderate	moderate	low	low	low	moderate	low	low
Dauty, M et al. [68] (2022)	severe	severe	low	moderate	moderate	low	low	moderate
de Geofroy, B et al. [69] (2024)	moderate	moderate	moderate	low	low	low	low	moderate
Dombrowski, N et al. [70] (2024)	moderate	moderate	low	low	moderate	low	moderate	moderate
Ebert, J et al. [71] (2018)	low	low	low	low	low	low	low	low

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Table 1 (*continued*)

Risk of Bias Assessment of Included Studies using the ROBINS-I Tool [13]								
Authors	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviation from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	Overall Bias
Elabd, O.M et al. [73] (2023)	moderate	low	low	low	low	moderate	moderate	low
Elabd, O.M et al. [72] (2024)	moderate	low	low	low	low	moderate	low	low
Fibiger, W et al. [17] (2014)	moderate	moderate	low	low	moderate	low	moderate	moderate
Flannery, S.W. et al. [74] (2023)	moderate	moderate	low	low	low	low	low	low
Fones, L et al. [75] (2021)	severe	moderate	moderate	low	low	low	low	severe
Franck, F et al. [76] (2021)	severe	low	moderate	low	moderate	low	low	severe
Garrison, J et al. [35] (2015)	moderate	moderate	low	low	moderate	moderate	low	moderate
Gokeler, A et al. [18] (2017)	severe	moderate	low	low	moderate	moderate	low	severe
Gonnachon, A et al. [77] (2024)	moderate	low	low	low	moderate	low	low	low
Guzel, N et al. [78] (2023)	moderate	moderate	low	low	low	low	low	low
Hamrin Senorski, E et al. [41] (2017)	severe	moderate	low	low	low	low	low	severe
Hebert, L et al. [19] (2011)	moderate	moderate	low	low	low	low	low	moderate
Hebert, L et al. [20] (2015)	moderate	moderate	low	low	moderate	low	low	moderate
Hunnicutt, J et al. [42] (2020)	severe	low	low	low	moderate	low	low	severe
Isaji, Y et al. [79] (2023)	moderate	low	low	low	moderate	moderate	low	low
Ishoi, L et al. [80] (2021)	low	low	low	low	low	low	low	low
Ithurburn, M et al. [81] (2021)	severe	moderate	low	low	moderate	low	low	severe
Ithurburn, M et al. [32] (2015)	severe	moderate	low	low	moderate	low	moderate	severe
Jack, R.A. et al. [82] (2023)	severe	moderate	moderate	low	moderate	moderate	moderate	moderate
Jang, S et al. [83] (2014)	low	low	low	low	low	moderate	low	moderate
Jimenez, A.E. et al. [84] (2021)	low	moderate	low	low	moderate	moderate	moderate	moderate
Johnston, P.T. et al. [85] (2022)	low	moderate	low	low	low	low	low	low
Kaur, M et al. [86] (2024)	low	low	low	low	low	moderate	low	low
Kehribar, L et al. [87] (2022)	moderate	low	low	low	low	moderate	moderate	low

Table 1 (*continued*)

Risk of Bias Assessment of Included Studies using the ROBINS-I Tool [13]								
Authors	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviation from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	Overall Bias
Kempfert, D.J. et al. [88] (2023)	low	low	low	low	low	low	low	low
Kew, M.E. et al. [89] (2022)	moderate	moderate	low	low	moderate	moderate	moderate	moderate
Kim, K et al. [21] (2013)	low	moderate	low	low	low	moderate	low	moderate
Kim, J.S. et al. [90] (2023)	moderate	moderate	low	low	moderate	low	moderate	moderate
Kitaguchi, T et al. [91] (2020)	moderate	moderate	moderate	low	low	low	low	moderate
Kivlan, B et al. [46] (2016)	moderate	moderate	low	low	low	low	low	moderate
Kivlan, B et al. [47] (2013)	moderate	moderate	low	low	low	low	low	moderate
Klasan, A et al. [92] (2021)	moderate	low	moderate	low	moderate	low	low	moderate
Kline, P et al. [93] (2015)	severe	moderate	moderate	low	low	low	moderate	severe
Kotsifaki, R et al. [94] (2023)	low	low	low	low	low	moderate	low	low
Lee, D et al. [29] (2018)	moderate	moderate	moderate	low	low	low	low	moderate
Lee, D et al. [95] (2019)	low	low	low	low	low	low	low	low
Lesnak, J et al. [22] (2019)	moderate	moderate	low	low	low	low	low	moderate
Losciale, J et al. [96] (2021)	severe	moderate	low	low	moderate	low	low	severe
Lunsford, B et al. [52] (1995)	low	low	low	low	low	low	low	low
Magill, J et al. [97] (2021)	low	low	low	low	low	low	low	low
Malloy, P et al. [49] (2016)	low	moderate	low	low	low	low	low	moderate
Meierbachtol, A et al. [98] (2021)	severe	moderate	low	low	low	low	low	severe
Moler, C et al. [99] (2024)	moderate	moderate	moderate	moderate	low	moderate	low	moderate
Monahan, P et al. [45] (2021)	moderate	moderate	low	low	low	low	moderate	moderate
Moran, T.E. et al. [100] (2023)	Moderate	Moderate	Low	Low	Low	Low	Moderate	Low
Moya-Angeler, J et al. [101] (2017)	Low	moderate	low	low	moderate	low	low	moderate
Myers, M et al. [36] (2018)	moderate	moderate	moderate	low	low	low	low	moderate
Nagai, T et al. [102] (2019)	severe	moderate	low	low	moderate	moderate	moderate	severe
Nawasreh, Z et al. [103] (2016)	moderate	moderate	moderate	low	moderate	low	low	moderate

(*continued on next page*)

Table 1 (*continued*)

Risk of Bias Assessment of Included Studies using the ROBINS-I Tool [13]								
Authors	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviation from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	Overall Bias
Niederer, D et al. [104] (2024)	moderate	low	low	low	low	low	low	low
Niederer, D et al. [105] (2024)	moderate	low	low	low	moderate	low	low	low
Norouzi, S et al. [106] (2019)	severe	moderate	low	low	low	low	low	severe
Norte, G et al. [107] (2020)	moderate	moderate	low	low	moderate	low	low	moderate
Norte, G.E. et al. [107] (2021)	moderate	moderate	low	low	low	moderate	low	low
Nyland, J et al. [108] (2020)	moderate	moderate	low	low	low	moderate	low	moderate
Ohji, S et al. [109] (2021)	severe	moderate	moderate	low	moderate	low	moderate	severe
Ong, M.T. et al. [110] (2024)	moderate	moderate	moderate	low	moderate	moderate	low	moderate
Paterno, M et al. [111] (2021)	severe	moderate	low	low	moderate	low	moderate	severe
Pedersen, J.R. et al. [112] (2022)	moderate	moderate	low	low	severe	low	low	moderate
Queiroz, N.C. A. et al. [113] (2024)	moderate	moderate	low	low	low	low	low	low
Raoul, T et al. [40] (2018)	severe	moderate	moderate	low	low	low	low	severe
Read, P.J et al. [115] (2022)	low	low	moderate	low	low	low	low	low
Read, P.J et al. [114] (2023)	low	severe	low	low	moderate	moderate	low	moderate
Rocchi, J et al. [116] (2020)	severe	moderate	moderate	low	moderate	low	moderate	severe
Roe, C et al. [117] (2020)	severe	moderate	moderate	low	moderate	moderate	moderate	severe
Ronden, A.E. et al. [118] (2022)	moderate	low	low	low	low	moderate	low	low
Schilling, D et al. [27] (2020)	severe	moderate	moderate	low	low	low	low	severe
Simon, J et al. [119] (2020)	low	moderate	low	low	low	low	low	moderate
Strong, A et al. [120] (2024)	low	moderate	low	low	low	moderate	moderate	low
Sueyoshi, T et al. [28] (2017)	low	moderate	low	low	low	moderate	moderate	moderate

Table 1 (*continued*)

Risk of Bias Assessment of Included Studies using the ROBINS-I Tool [13]								
Authors	Bias due to confounding	Bias in selection of participants into the study	Bias in classification of interventions	Bias due to deviation from intended interventions	Bias due to missing data	Bias in measurement of outcomes	Bias in selection of the reported result	Overall Bias
Suits, W.H. et al. [121] (2024)	severe	moderate	moderate	low	moderate	moderate	low	moderate
Swindell, H et al. [122] (2023)	moderate	severe	moderate	low	low	moderate	moderate	moderate
Tavares, M.L. A. et al. [123] (2023)	moderate	low	low	low	moderate	moderate	moderate	moderate
Teichmann, J et al. [124] (2022)	low	low	low	low	low	low	low	low
Thompson, X. D. et al. [125] (2022)	low	low	low	low	low	low	low	low
Toale, J et al. [126] (2021)	severe	moderate	low	low	moderate	low	moderate	severe
Ueda, Y et al. [127] (2022)	moderate	low	moderate	low	low	low	low	low
VanZile, A et al. [128] (2022)	low	low	low	low	low	low	low	low
Wang, A et al. [129] (2024)	low	moderate	low	low	moderate	low	low	low
Weber, M et al. [130] (2024)	moderate	low	low	low	moderate	low	low	low
Webster, K et al. [131] (2020)	severe	low	moderate	low	low	moderate	low	severe
Welling, W et al. [132] (2020)	severe	moderate	moderate	low	low	moderate	low	severe
Welling, W et al. [30] (2018)	severe	moderate	moderate	low	moderate	moderate	low	severe
Werner, J et al. [37] (2018)	severe	moderate	moderate	low	low	moderate	low	severe
White, A et al. [31] (2018)	low	moderate	low	low	low	low	low	moderate
Worner, T et al. [48] (2019)	moderate	moderate	low	low	low	moderate	low	moderate
Xergia, S et al. [133] (2014)	low	moderate	low	low	low	low	low	moderate
Zarro, M et al. [134] (2023)	low	low	low	low	low	low	low	low
Zhou, L et al. [135] (2022)	low	moderate	low	low	low	low	low	low
Zwolski, C et al. [136] (2016)	moderate	moderate	low	low	low	low	low	moderate

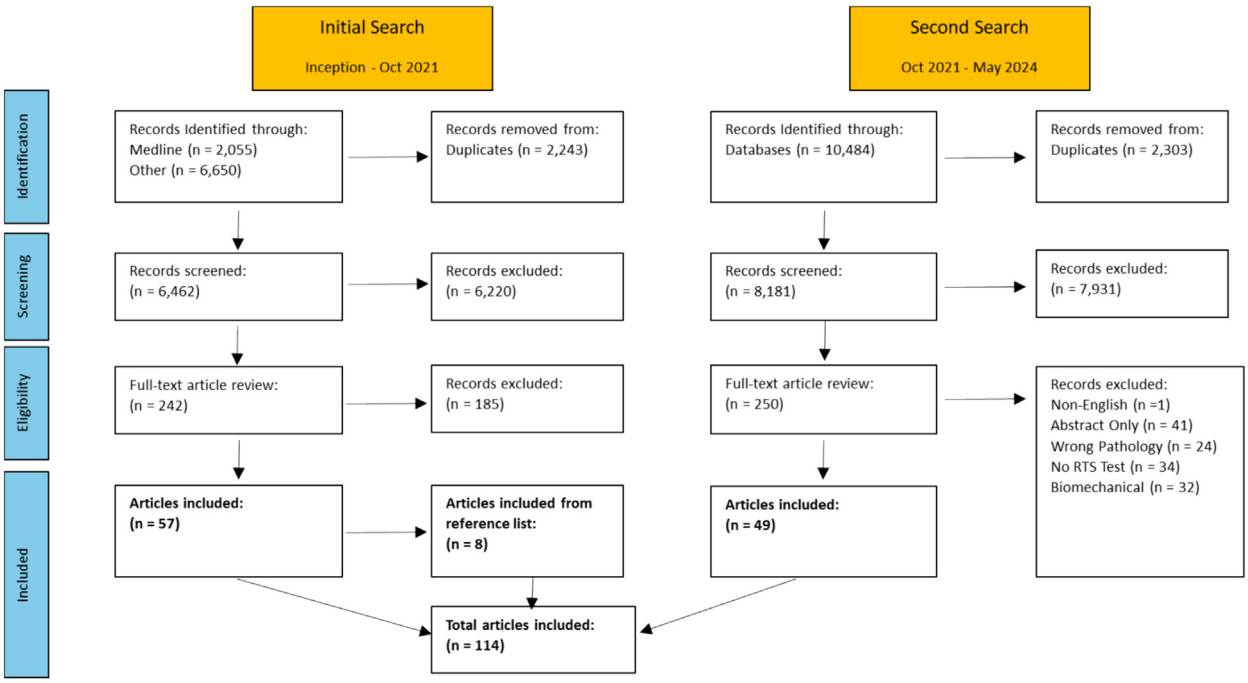

Figure 1. Describes the flow diagram for the study's inclusion and exclusion process. Two separate searches were performed to gather all articles from inception to MAy 2024.

used to meet inclusion criteria included: femoroacetabular impingement and labral repair, while ankle injuries included: lateral ankle reconstruction/Bostrom, syndesmotic repair, Jones fracture, Lisfranc injury, and ankle fracture. Studies were excluded if any of the following were present: arthroplasty, prior lower extremity injury over the year prior, average age younger than 14 or older than 40 years old, or no evaluation of a return to sport test or functional test.

2.3. Data extraction/analysis

Initial exclusion of articles was performed through the removal of duplicates via EndNote and Rayyan. Titles and abstracts were screened for relevance by two authors. The manuscripts that were included at this point in time were then obtained for full text analysis. The same 2 authors reviewed the full texts to assess for eligibility using the criteria stated below. If any discrepancies arose during inclusion and exclusion, a senior author was consulted for the final inclusion and exclusion decision. To assess bias for the included studies, the authors utilized the ROBINS-I Tool and illustrated the results in Table 1 [13].

3. Results

3.1. Literature search

8,705 studies were identified through the initial search and 10,484 articles were identified in the second search, yielding 19,189 articles. This was then narrowed down to 14,643 after removing duplicates. 14,151 additional studies were then removed based on inclusion/exclusion criteria, leaving 492 studies for review. The full texts for all 492 articles were reviewed and assessed for eligibility criteria. 106 of the studies met all inclusion criteria and were included in the systematic review. An additional 8 studies were included in the systematic review that were found by reviewing the references of the included studies (114/19,189; 0.6%). Figure 1 denotes the flow diagram used for the study.

3.2. Study quality, bias, and characteristics

As depicted in Table 1, the majority of study's overall bias was identified as moderate to severe with significant concerns for potential confounding bias as well as bias in selection of study participants. The included studies were then evaluated for the outcomes listed above and primarily incorporated into Table 2.

Table 2
Manuscript Patient Demographics and Criteria for Return to Sport.

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Aizawa J, et al. [55] (2022)	54	20.0	21/33	ACLR	ACL-RSI, SLH, BDX-4, Y balance,	increased knee flexion strength, H/Q, leg anterior reach distance LSI, and lateral and medial SLH are important to excess the ACL-RSI for a RTS
Aizawa J, et al. [56] (2024)	59	20.0	24/35	ACLR	ACL-RSI, Biodex Multi-Joint Testing and Rehabilitation System torque metre, H/Q, Y Balance, SLH	higher running ability and limb symmetry in the medial SLH are positively associated with psychological readiness
Albano, T et al. [38] (2020)	150	RTS: 28.7 ± 7.1 Not RTS: 27.4 ± 7.2	RTS: 91.9 Not RTS: 79.7	ACLR	Balance system SD dynamometer, *iso*-K dynamometer, ACL-RSI, IKDC	Athletes with increased quad strength tended to RTS at any level more quickly; psychological readiness was the most important factor in not returning followed by a better agonist: antagonist ratio and better balance
Alswat, M et al. [57] (2020)	93	35.2 ± 6.7	98.9	ACLR	IKDC, TSK-11	The participants in this study had a low rate of return with fear of reinjury being the most common reason not to return. However, a participant's IKDC and TSK-11 scores were associated factors for RTS, thus optimizing those factors after surgery is crucial.
Alzakerin, H et al. [58] (2020)	62	Post-ACLR: 20.4 + 6.2 Control: 20.9 + 3.4	Post-ACLR: 51.6 Control: 56.2	ACLR	Tegner Activity Scale, Running speed	No difference in Tegner score. A strong first-order relationship between the peak vGRF and the linear vGRF loading rate was observed in the control right, control left, and post-ACLR nonreconstructed limbs, whereas a weaker relationship was observed in the reconstructed limb of post-ACLR individuals. The normal probability plot of the residuals that the first-order model was appropriate: linear vGRF loading rate = 0.06 + 7:90 x peak vGRF
Aujla R.S., et al. [59] (2021)	196	26.8 ± 9.5, 27.5 ± 8.6	123/73	ACLR	PROM (IKDC, KOOS, Lysholm, Tegner, Noyes, Subjective Knee Eval Form), KT-1000 arthrometer, SLH for distance, 6-m timed hop, Triple hop for distance, triple crossover hop for distance, *iso*-dynamommeter	Patients with AUG-ACLR had higher 1-year postoperative Tegner scores and rates of RTS and preoperative sport level compared with the HA-ACLR group
Birchmeier, T et al. [25] (2019)	52	22.9 + 5.0	36.5	ACLR (8 had more than one ACLR)	SL drop vertical for max height, SLHD, THD, *iso*-M Q strength	SLHD may indicate improved knee extension strength; THD may indicate improvement reactive strength
Bodkin, S et al. [60] (2017)	51	22.9 + 4.5	29.4	ACLR	*iso*-K Q/H strength, SLHD, THD, CHD, 6-m timed hop, KOOS, IKDC	Relationship between muscle function and subjective knee function differ between different cohorts organized by time out of surgery suggesting that time from surgery still needs to be considered as a factor for RTS

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Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Bodkin, S et al. [34] (2018)	108	21.7 + 8.3	54.6	ACLR	IKDC, KOOS, SL balance	SL balance not predictive enough in a straight knee position to detect impairments and does not predict subjective function in ACLR patients
Borawski J, et al. [61] (2024)	65	20.6 ± 1.8 (17–27)	35/30	ACLR	Jump Landing task, Knee JPS, iso dyna, LSI, ACL-RSI, TSK-11	ACL-RSI score was significantly related to LSI peak quad tourque.
Browning, R et al. [51] (2021)	58	31.9 ± 12.2	31.0	Hip Arthroscopy	HOS-SS, PCS, TSK-11	Patient kinesiophobia and pain catastrophizing at 1-year follow-up were negatively correlated with RTS and achievement of a CSO in sport-related activities at 2-year follow-up.
Brunst, C et al. [62] (2020)	27	na	18.5	ACLR	Iso-M Q	Early analysis indicates that achieving quadriceps femoris strength symmetry ≥90% before the release to sports may mitigate future degenerative changes in the knee cartilage matrix
Burland, J et al. [39] (2018)	51	15.9 + 1.8	54.0	ACLR	*iso*-K and *iso*-M Q/H, ACL-RSI, Pedi-IKDC	Higher strength measures at 3/6 months post-op was associated with greater self-reported knee function and greater readiness to RTS
Burnham, J et al. [63] (2016)	71	25.5 + 0.6	62.3	ACLR	Hip abd./ER/extension peak *iso*-M force, trunk endurance (plank) relationship with SL step down test	Hip/trunk muscle function correlates strongly with SL Step down performance (especially in females)
Byrne, L et al. [64] (2021)	control: 155 meniscectomy: 128 repair: 30	control:23.33 ± 5.0 meniscectomy: 22 ± 5.4 repair: 23.75 ± 5.6	100	ACLR and Meniscectomy, meniscal repair, no meniscal intervention	PROs (Marx Activity Rating Scale, the ACL-RSI, IKDC) Iso-K Q/H, countermovement jump, drop jump	No significant between-group differences were identified in any metric relating to patient-reported outcome measures (p = 0.611), strength and jump measures (p = 0.411) or the ability to achieve symmetry-based return to play criteria (p = 0.575). Clinically, these results suggest that concomitant meniscal surgery has no significant effects on patient-reported outcome measures, strength and jump metrics at the return to play stage post-operatively and can inform the pre-operative counselling of those awaiting ACL reconstruction surgery with likely meniscal intervention
Caffrey, E et al. [53] (2009)	Unilateral ankle instability (UAI): 30 Control: 30	UAI: 20.5 ± 1.6 Control: 20.0 ± 1.0	UAI: 50.0 Control: 50.0	Functional ankle instability	Timed figure-of-8 hop, timed, side hop, timed 6 m crossover hop, and timed square hop	Individuals who experience subjective unilateral ankle instability and who reported instability during actual functional performance had a significantly slower times in 3 of the 4 hop tests when compared to healthy participants. These results support the hypothesis that performance deficits exist in individuals with ankle instability when using single-limb functional tests.

Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Cristiani, R et al. [65] (2019)	2335	29.7 ± 10.9	50.6	ACLR	*iso*-K Q strength, SLHD, PASS, KOOS	A symmetrical 6-month *iso*-k quadriceps strength and SLHD consistently increased opportunity for PASS (patient-acceptable symptom state)
Cronstrom A, et al. [66] (2023)	143	25.0 ± 5.7	72/71	ACLR	PROMs(KOOS, ACL-RSI), Ankle and Hip ROM, Handheld dynamometer, isokinetic dynamometer, SLHD, side hop	Lower LSI was associated with worse KOOS Sport and Rec subscore. Shorter Hop distance associated with lower ACL-RSI score
Crotty N.M.N., et al. [67] (2022)	116	24.3 ± 6.5	male	ACLR	SLDJ, isokinetic concentric knee extensor peak force, RSI, IKDC	Isokinetic knee extension strength explained 30% of SLDJ performance, with weaker relationship between knee flexion strength and SLDJ performance. Iso strength and SLDJ performance were weak predictors of variation in IKDC scores
Dauty M, et al. [68] (2022)	539	25 ± 7	155/384	ACLR with and without arthrofibrosis	iso dyna LSI, Lysholm, Tegner	Quad and Hamstring LSI remained low 12 months after surgery and RTS was low in arthrogenic ACLR. Few arthrogenic pts RTS at 7 months post-op
de Geofroy B, et al. [69] (2024)	144	27.6	mostly male	ACLR	isokinetic testing, military aerobic fitness tests	Return to same level of specific annual fitness tests in the French Army is 40% after ACLR
Dombrowski N, et al. [70] (2024)	127	M – 20.14 ± 6.9, F – 19.13 ± 8.51	64/63	ACLR	ACL-RSI, hand dyna, Y-balance, single-leg vertical jump, single leg hop w/ fatigue protocol, single leg crossover hop, single leg Figure 8 hop, single leg drop landing knee excursion, leg press reps to failure	Male ACL-RSI can be predicted using only one physical function outcome while no physical function variables predicted ACL-RSI in women. Knee excursion was the only contributor to ACL-RSI in men. No relationship between physical test and ACL-RSI in women.
Ebert, J et al. [71] (2018)	40	23.8 (15–43)	62.5	ACLR	6-m timed hop, SLHD, THD, CHD, *iso*-K Q/H, Back In Action (BIA) battery	BIA test more challenging than typical four-hop test battery and isokinetic testing
Elabd O.M,, et al. [73] (2023)	50	Experimental- 22.84 ± 3.87, Control- 23.72 ± 3.56	male	ACLR	VAS, Verticle jump, hop for distance, drop jump followed by double hop for distance, square hop, side hop, KOOS, knee effusion	Criterion based rehab protocol proved to improve all functional outcomes as well as KOOS and knee effusion
Elabd O.M,, et al. [72] (2024)	100	Experimental- 21.91 ± 1.74, Control- 22.09 ± 1.83	male	ACLR	VAS, KOOS, knee effusion, verticle jump, hop for distance, drop jump followed by double hop for distance, square hop, side hop	Accelerated rehab protocol is more effective in improving functional outcomes than a conventional rehab protocol in amateur atheletes with ACLR
Fibiger, W et al. [17] (2014)	42	22.2 ± 3.6	45.2	ACLR	Ultrasound (swelling), *iso*-K Q/H	Suggested criteria for RTS: quadriceps strength 85% LSI, hamstring strength 90% LSI, HQ ratio of > 70% with use of isokinetic testing

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Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Flannery S.W., et al. [74] (2023)	65	19.4 ± 5.1	27/38	ACLR	SLH for distance, arthrometer for knee laxity, IKDC	CSA, volume, and estimated failure load at 6 months were predictive of change in single leg hop ratio
Fones, L et al. [75] (2021)	74	15.9 ± 1.5	37.8	ACLR	ACL-RSI, IKDC	Both knee function and psychological responses to injury were important in determining an adolescent athlete's return to sport. The findings support the use of the IKDC and ACL-RSI at midterm follow-up, with higher scores associated with a greater likelihood of adolescent patients returning to sport after ACLR.
Franck, F et al. [76] (2021)	676	27.6 ± 10.4	70.4	ACLR	K-STARTS (ACL-RSI, QASLS, SLHD, THD, Side Hop Test, CHD, MICODT)	Completion of a specific RTS program, in addition to standard rehabilitation, was the most important factor influencing the K-STARTS composite functional and psychological RTS test score at 6 months after ACL reconstruction
Garrison, J et al. [35] (2015)	40	17.2 ± 3.8	50.0	ACLR	THD, SLHD, *iso*-M Q strength, Vail Sport test, YBT	>4 cm difference in anterior reach at 12 weeks did not tend to achieve 90% LSI for SLHD and THD at RTS
Gokeler, A et al. [18] (2017)	28	M: 26.2 ± 8.7 F: 22.4 ± 6.4	78.6	ACLR	*iso*-K Q/H strength, SLHD, THD, side hop, LESS, IKDC, ACL-RSI	Majority of patients are not ready to RTS at 6-month post-op with suggested battery: LSI > 90% on quad and hamstring *iso*-K strength; H/Q ratio > 55% for F and > 62.5% for males; LESS < 5; LSI > 90% for all hoping tasks; ACL-RSI > 56 points; IKDC score within 15th percentile of healthy gender-age-matched subjects
Gonnachon A, et al. [77] (2024)	83	24 +- 7.2	67/16	ACLR ± ALLR	isokinetic dynamometer, flexor/extensor ratios, ACL-RSI	Do difference in isokinetic measurements or ACL-RSI scores between isolated ACLR and ACLR with ALLR. Combined ACLR with ALLR is a viable option for pivoting athletes at risk of rerupture.
Guzel N, et al. [78] (2023)	22	28.32 ± 6.6	Male	ACLR	Single hop for distance, triple hop for distance, crossover triple hop for distance, medial side triple hop for distance, medial rotation with hop for distance, Y-balance, Lysholm, Tegner, IKDC	6 month SLHT were lower in ACLR side compared to healthy. Similarities in LSIs in SLHTs and balance scores of ACL and healthy sides reveals MAI technique can be used in athletes from a functional point of view
Hamrin Senorski, E et al. [41] (2017)	157	15–30	51.0	ACLR	Tegner activity scale; *iso*-T Q/H strength, SLHD, vertical jump, side hop	Patients who RTS after ACLR have better subjective knee function and higher self-efficacy of knee function; further emphasis should be placed on psychological factors
Hebert, L et al [19] (2011)	74	4–17.5	50.0	healthy	Iso-M	For the adolescent subgroup, the majority of muscle groups test (with the exception of ankle dorsiflexors); maximal isometric torque (MIT) values showed acceptable intra- and interrater reliability and concurrent validity with the Cybex

Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Hebert, L et al. [20] (2015)	351	4–17.5	50.0	healthy	Iso-M	Provides HHD reference values that may be helpful in pediatric population especially when bilateral impairments are present
Hunnicutt, J et al. [42] (2020)	30	21.5 (14–41)	63.3	ACLR	Knee extensor *iso*-M and *iso*-K peak torque, cross sectional area each quad with MRI, SLHD, step length, IKDC, KOOS	Greater symmetry in size/ strength of quadriceps (not activation) correlated with physical performance, greater symmetry in strength was more strongly correlated with patient related function
Isaji Y, et al. [79] (2023)	66	17.3 ± 2.6	17/49	ACLR	TSK-11, SLHT, handheld dynamometer, AKPS, KS measure	Kinesiophobia associated with low anterior knee pain score and low SLH test 6 months after ACLR. Pts w/ low SLH score or severe pain may need rehab to reduce kinesiophobia
Ishoi L, et al. [80] (2021)	45	30.6 ± 5.9	34/11	FAIS	handheld dynamometer for isometric hip force, drop jump test for single leg reactive strength index, 5SST, HAGOS questionnaire	Higher maximal hip extension strength associated with greater ability to participate fully in preinjury sport after hip arthroscopy for FAIS
Ithurburn, M et al. [81] (2021)	57	17.6 ± 3.0	75.4	ACLR	Iso-M Q, DVJ landing	Young athletes after ACLR with quadriceps strength asymmetry at the time of RTS favored the uninvolved limb during DVJ landing 2 years later. These landing asymmetries may relate to long-term knee joint health after ACLR.
Ithurburn, M et al. [32] (2015)	93	17.4	30.1	ACLR	SL drop landing	At time of RTS, athletes who had undergone ACL reconstruction demonstrated asymmetry during a single-leg drop landing task compared to healthy controls; negative correlation with lower quad strength increasing asymmetry
Jack R.A., et al. [82] (2023)	32	control-24.1 ± 7.2 BFR-25.2 ± 2.8	19/13	ACLR	single leg squat, single leg eccentric step down, Y-balance, single leg leg press, single leg hamstring curl	BFR may decrease muscle mass and bone loss for 12 weeks post-op and improve time to RTS with functional outcomes comparable with standard rehab
Jang, S et al. [83] (2014)	67	21.9 + 4.0	–	ACLR	IKDC, Lysholm score, KT-2000 arthrometer, SLHD, co-contraction test, carioca test, shuttle run test	Tests for rotational stability (carioca and co-contraction) were the only statistically significant tests between RTS and non-RTS groups
Jimenez A.E., et al. [84] (2021)	19	22.6	15/4	irreperable hip labral tear	mHHS, HOS-SSS, NAHS	Primary arthroscopic labral recon for irreperable labral tears was associated with significant improvement in PROs and high rates of RTS in high level athletes.
Johnston P.T., et al. [85] (2022)	QT-35, HT-70	QT-20 (15–34), HT-20(15–32)	82/23	ACLR	PROMs(KOOS-QOL, IKDC, ACL-RSI, Marx), KT-1000 arthrometer, single hop for distance, triple crossover hop for distance, HUMAC NORM dynamometer	hop testing demonstrated LSI > 90% at 6 months after ACL recon. QT group had lower knee extensor strength symmetry compared to HT.

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Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Kaur M, et al. [86] (2024)	245	competative-20.1 (4.4), rec-27.1(6.6)	N/A	ACLR	TSK-17, ACL-RSI, IKDC, KOOS, isokinetic dynamometer,	Competative athletes had higher scores for IKDC and quad peak torque as well as quad strength. Psychological measures did not differ between groups
Kehribar L, et al. [87] (2022)	39	25.8 (6.6)	male	ACLR	Lysholm, Tegner, IKDC, SLHD, THD, CHD, medial side triple hop for distance, medial rotation hop for distance	similar LSI rates of the SLHTs applied in different directions in the ST/G and MAI techniques assumes that MAI can get athletes back to RTS criteria adequately
Kempfert D.J., et al. [88] (2023)	12	26.5 ± 3.5	4 m, 8f	ACLR	adapted crossover hop test for distance	This study found the ACHT to have excellent intra and inter rater reliability
Kew M.E., et al. [89] (2022)	100	16.1 ± 1.4	40/60	ACLR	PROMs(IKDC, KOOS, TSK), isokinetic, concentric knee extension and flexion strength (IV dynamometer), SLH for max distance, Triple hop for distance, triple crossover for distance, 6 m timed hop	Pts with ACLR graft failure had greater and more symmetric quad strength at time of RTS compared with pts with no secondary injury.
Kew M.E., et al. (2022)	100	16.1 ± 1.4	40/60	ACLR	PROMs(IKDC, KOOS, TSK), isokinetic, concentric knee extension and flexion strength (IV dynamometer), SLH for max distance, Triple hop for distance, triple crossover for distance, 6 m timed hop	Pts with ACLR graft failure had greater and more symmetric quad strength at time of RTS compared with pts with no secondary injury.
Kim, K et al. [21] (2013)	27	23.9 + 2.9	0	healthy	Iso-M knee extensor strength (fixed and non-fixed) measured against Iso-K	Fixed HHD and isokinetic dynamometry method were statistically significant. Fixation of HHD during measurement in the supine position increases the reliability and validity in measuring the quadriceps strength
Kim J.S., et al. [90] (2023)	48	28.9 ± 8.3	35/13	ACLR	KT-2000 arhtrometer, BBS-APSI, Tegner, IKDC, Lysholm, single leg hop, single leg verticle jump, isokinetic dynamometer	BBS-APSI correlated with functional performance test scores after ACLR rendering BBS-APSI as a useful tool to evaluate postop functional recovery.
Kitaguchi, T et al. [91] (2020)	124	17.0 ± 2.7	40.3	ACLR	SLHD	Functional single leg hop test for RTS group was significantly higher than non-RTS group.

Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Kivlan, B et al. [46] (2016)	Femoroacetabular impingement (FA): 15 Control: 13	FA: 19.6 ± 1.4 Control: 18.9 ± 1.3	FA: 0 Control: 0	Femoroacetabular impingement	Passive range of motion for hip flexion, external rotation and internal rotation, isometric hip flexion, extension, abduction, adduction, internal rotation and external rotation strength, medial triple hop test for distance, lateral triple hop test for distance, 6 m crossover hop test for time	Medial triple hop and lateral triple hop tests should be used in assessment of dancers with suspected femoroacetabular impingement (FAI). FAI dancers had less hip extensor strength and performed worse on the medial and lateral hop test. Range of motion did not appear to be affected.
Kivlan, B et al. [47] (2013)	19	18.9 ± 1.1	0	Unilateral hip pain	Crossover reach test for distance, lateral triple hop test for distance, medial triple hop test for distance, 6 m crossover hop test for time	Functional performance tests in this study can be reliably performed on dancers with unilateral hip pain. The medial hop test was the only functional performance with evidence of validity in side-to-side comparisons meaning it may be. A reliable and valid performance test to assess impairments related to hip pain among dancers.
Klasan, A et al. [92] (2021)	123	30 (18–60)	57.7	ACLR	Tegner Activity Scale, ACL-RSI, KT1000, Lysholm, IKDC	Preoperative Tegner and expected Tegner level collected prior to an ACL reconstruction can aid in the objective prediction of patients' return to sport after 2 years. High-level athletes are more likely to return to their previous sport and to the previous level. Younger patients achieve a higher Tegner level at 2 years.
Kline, P et al. [93] (2015)	30	21.3 + 7.6	46.7	ACLR	*iso*-M knee quad strength, YBT-ant., SL Step down	Better performance SL stepdown at 3 months correlates with improved sagittal plan knee mechanics during running at 6 months post-op
Kotsifaki R, et al. [94] (2023)	658	ACLR-24 ± 6, Control-25 ± 5	male	ACLR	double countermovement jump, single leg countermovement jump, double leg drop jump, single leg drop jump	despite passing traditional discharge criteria at RTS, athletes remained asymmetrical during vertical jumps in the concentric phase, landing phase from bilateral jumps, and most performance metrics.
Lee, D et al. [29] (2018)	75	27.5 + 9.2	100	ACLR	IKDC, Lysholm score, Tegner activity scale, ACL-RSI, KT-2000 arthrometer, *iso*-K Q/H strength, SLVJ, SLHD, co-contraction test, carioca test, shuttle run test	Significant correlations between SLVJ and subjective IKDC, Tegner, ARL-RSI, *iso*-K extensor muscle and all functional tests; SLVJ valid for RTS test
Lee, D et al. [95] (2019)	52	25.8 + 5.0	90.4	PCLR	Lysolm score, Tenger activity scale, IKDC, CT, MRI, stress radiography, *iso*-K Q/H strength, SLHD, SLVJ	Extensor deficit, flexor deficit, and patient satisfaction were significantly associated with failure of RTS

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Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Lesnak, J et al. [22] (2019)	20	23.7 + 2.0	60.0	healthy	Iso-M Q with HHD, Iso-K Q	Results show it is possible to obtain valid measures of quadriceps peak torque and late rate of torque development (RTD) using an HHD. Measures of early RTS and RTS avergae obtained with an HHD were more variable and should be viewed with caution.
Losciale, J et al. [96] (2021)	205	RTS-passing: 15.4 ± 2.4 RTS-fail: 17.5 ± 3.4	30.7	ACLR	Iso-M Q, SLHD, THD, CHD 6-m timed hop, IKDC	ACLR participants were categorized as passing RTS criteria (RTS:PASS group) if they met 90% LSI on quadriceps strength and each of the hop tests, as well as scored 90 on the IKDC. Failing RTS criteria (RTS:FAIL group) occurred if at least one criterion was not met.
Lunsford, B et al. [52] (1995)	203	20–59	60.1	Healthy	Standing heel raises	The average number of heel-rises was 27.9 (SD = 11.1, minimum = 6, maximum = 70) for all groups and both genders, with no differences between male and female subjects. The lower 99% confidence interval was 25. A recommendation is made to change the standard of testing plantar-flexion function, when using the standing heel-rise test, to require 25 repetitions for a grade of Normal.
Magill, J et al. [97] (2021)	100	11.7 ± 3.6	48	healthy	Stork test, stork test on Bosu, single-leg squat, single-leg squat on Bosu, Quadrant hop-clockwise, quadrant hop counterclockwise, SLHD, 6-m timed hop, CHD	Chronological age is a strong predictor of performance on these RTS tests allowing for the creation of growth curves that can help clinician benchmark their patients against healthy, uninjured pediatric normative group when making RTS decisions. Caution should be practiced when using LSI in isolation for RTS in pediatric athletes
Malloy, P et al. [49] (2016)	23	19.4 ± 0.8	0	Healthy	Maximal isometric hip strength for hip abductor and external rotator muscle groups, hip and knee excursion and peak hip and knee internal join moments in all planes for drop tests (single leg drop land and hold, single leg drop land and side cut, single leg drop land and forward run)	The greater hip external rotator strength was associated with greater peak hip external rotation and abduction moments, greater peak knee internal rotation moments, and greater hip frontal plane excursion during unanticipated single leg landing tasks. The study shows that the hip external rotator strength may play a role in the control of dynamic valgus during unanticipated landing and cutting. Therefore, practitioners should consider implementing hip muscle strengthening into training programs for landing and cutting athletes.

Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Meierbachtol, A et al. [98] (2021)	Training: 23 No Training: 23	Training: 19.2 + 5.4 No Training: 19.5 + 5.3	Tr: 39.1 No Tr: M: 39.1	ACLR	ACL-RSI, SLHD, THD, CHD, and 6-m timed hop	Return to sport criteria were ACL-RSI score ≥ 70 points and limb symmetry index ≥ 90% on all hop tests. At follow-up, KASE score was higher in TRAINING than No TRAINING (92.7 vs. 89.1 points; respectively), but ACL-RSI, TSK-11 and fear intensity scores were not significantly different between groups.
Moler C, et al. [99] (2024)	314	19.7(4.8)	146/168	ACLR	PROs(IKDC), single leg hop test, isokinetic dynamometer	following ACLR with bone patellar tendon bone grafts, hamstring function should be more closely monitored for optimal short term recovery
Monahan, P et al. [45] (2021)	RTS: 83 No RTS: 53	20.8 ± 7.1	RTS: 38.6 No RTS: 24.5	Hip Arthroscopy	Alpha angle, modified Harris Hip Score, Hip Outcome Score-Sport Specific-Subscale, and Nonarthritic Hip Score	Postoperative alpha angle was identified as a predictor of return to sport in athletes. The odds of returning to sport were 6.3 times greater in athletes with postoperative alpha angles 46 compared with athletes with angles > 46
Moran T.E., et al. [100] (2023)	96	cohort 1–19.6 ± 10.6, Cohort 2–29.5 ± 13.3	53/33	ACLR	isokinetic dynamometer, single leg hop for distance, triple hop for distance, 6-m hop for time, IKDC, KOOS	no differences in LSI or contralateral limb functional performance between pts undergoing contralateral HS autograft for augmentation of smaller diameter HS autografts harvested from injured extremity
Moya-Angeler, J et al. [101] (2017)	74	34.0 ± 9.0	100	ACLR	SLHD, drop vertical jump, vertical jump, gait and sprint	Increased symmetry restoration in kinetics of involved and uninvolved limb up to 12 months post-op; importance of isolating single limb in testing procedures
Myers, M et al. [36] (2018)	45	21.1 ± 5.8	64.4	ACLR	YBT, *iso*-K PKET, PKFT and HQ ratio	YBT and isokinetic strength testing can expose asymmetries and impact RTS decision making
Nagai, T et al. [102] (2019)	26	22.1 ± 9.1	42.3	ACLR	SLHD, THD, unilateral peak leg power (leg press,) *iso*-K Q/H	Peak knee extension torque positively associated with hop distance; encouraged use of isokinetic testing with RTS as they provide significantly lower LSI values than hop testing
Nawasreh, Z et al. [103] (2016)	95	27.14 + 10.6	66.3	ACLR	RTAC (includes *iso*-M Q strength, SLHD, THD, CHD, 6-m times hop, KOS-ALDS, GRS)	Passing RTAC early after ACL more likely to demonstrate normal knee function and movement symmetry at 12 and 24 months post-operatively
Niederer D, et al. [104] (2024)	112	25.6(18–35)	53/47	ACLR	drop jump, balanced front hop test, balanced side hop test, front hop for distance, PROs (ACL-RSI, KOOS, Tegner, TSK)	Stop-X intervention was slightly superior to usual care as late stage rehab for ACLR.
Norouzi, S et al. [106] (2019)	27	23.8 + 3.3	100	ACLR	Bilateral jump landing task	Reduced hip abduction during contact phase of landing in ACLR athletes who have RTS; reduced hip abduction during jump landing is a risk factor for ACL re-injury

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Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Norte, G et al. [107] (2020)	80	21.7 + 7.1	52.5	ACLR	*iso*-K and *iso*-M Q/H strength, SLHD, THD, CHD, 6-m timed hop	Isokinetic quadriceps strength test most difficult to pass; SLHD was easiest – graft type had influence on results
Norte, G.E., et al. [107] (2021)	160	ACLR-21.7 ± 7.8, Control-22.1 ± 4.1	79/81	ACLR	isokinetic dynamometer, isokinetic H/Q ratio, single leg hop, triple hop, crossover hop, 6-m timed hop, IKDC	isokinetic quad strength was the most difficult test to pass, and SLHT were the easiest. Graft type, time for surgery, PA level, and meniscal procedure may influence ability to meet RTS criterion
Nyland, J et al. [108] (2020)	150	20.3 ± 7.2	55.3	ACLR	KOS-SAS (>85%), LSI ≥ 90% eight rep max single leg press, LSI ≥ 90% 2 out of 3 hop tests (THD, 20 m single leg timed hop, and 20 m single leg timed crossover hop)	Supplementing primary ACL reconstruction and standard physical therapy with a return to sports bridge program prior to release to unrestricted sports performance was effective at improving patient outcomes and decreasing ipsilateral knee re-injury and contralateral knee injury rates.
Ohji, S et al. [109] (2021)	50	20.5 ± 11.0	46	ACLR	ACL-RSI	Participants with low ACL-RSI scores 6 months post-ACLR had low preoperative ACL-RSI scores that did not improve preoperatively to 6 months post-ACLR
Ong M.T., et al. [110] (2024)	36	ECC-26.3 ± 6.7, control-25.6 ± 4.3	31/5	ACLR	isokinetic Q and H test, single leg hop for distance,	6 week course of eccentric isokinetic training after ACLR increased quad and hamstring strength more effectively than concentric training and resulted in better functional performance and thus a higher RTS ratio
Paterno, M et al. [111] (2021)	159	17.2 ± 2.6	29.6	ACLR	IKDC, KOOS-QOL, Iso-M Q, SLHD, THD, CHD, 6 m timed hop	Confident, young athletes (as measured by IKDC and KOOS-QOL) at the time of RTS after ACLR are more likely to meet al. l RTS criteria but also more likely to sustain a 2nd ACL injury after RTS
Pedersen J.R., et al. [112] (2022)	39	28 (24–35)	21/18	ACLR	KOOS, SLHD, CHD, side hop test, agility T test, quad and hamstring isometric MVC	3 months of public municipal rehab improves PROs and LE muscle strength in non-surgically treated patients with ACL injury, however, only 1 in every 10 pts passed RTS criteria
Queiroz N.C. A., et al. [113] (2024)	34	29.6 ± 6.8	26/8	ACLR	Y balance, SLH, functional movement screen, isokinetic dynamometer (OKC, CKC), IKDC	Self reported symptoms prevelance and lower limb symmetry were associated with knee extension strength and functional performance (YBT)
Read P.J et al. [115] (2022)	51	ACLR-22.5 ± 4, control-25.2 ± 4.4	male	ACLR	single leg drop jump-RSI, ground reaction force, Stretch shortening cycle	RSI changed significantly from initial assessment to follow up but should not be the sole metric as observed increases did not correspond with changes in SSC function
Read P.J et al. [114] (2023)	268	ACLR-24.2 ± 5.1, control-24.7 ± 4.3	male	ACLR	10-second hop test, RSI, single leg drop vertical jump RSI, isokinetic dynamometer, IKDC	SLVDJ, RSI, and knee-extension torque were significant predictors of 10-sec hop height and RSI. Vertical hop deficits were present after ACLR with rehab and should be included in RTS test battery

Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Raoul, T et al. [40] (2018)	234	28.4 ± 8.6	73.9	ACLR	*iso*-K Q/H strength, SLHD, THD, CHD, 6-m timed hop; ACL-RSI	Mean of 6 months S/P ACLR is generally unsatisfactory recovery and functional performance for RTS; ACL-RSI is significantly correlated with isokinetic and hop tests
Rocchi, J et al. [116] (2020)	15	na	100	ACLR	EMG, Initial Contact and Maximum postimpact knee angle with 4 single leg landing tasks (Stop landing, smooth landing, rebound landing, SLHD)	At the time of return to sport, ACL-R subjects showed altered motor control strategies of single-leg landings. These alterations may lead to uncoordinated movement, hence increasing the risk of reinjury
Roe, C et al. [117] (2020)	66	21.2 ± 6.9	47.0	ACLR	IKDC, Lysolm Activity Scale, TSK-11, KOOS, SL step down, YBT anterior	SL step down demonstrates stronger relationship to patient-reported knee function than YBT-A, SL step down symmetry strongly correlated with fear of movement
Ronden A.E., et al. [118] (2022)	103	PASSED-18.5(15), FAILED-21.0(7)	78/25	ACLR	Back in action test battery, PROMs (IKDC, KOOS, ACL-RSI)	low % of patients passed BIA test 9 montsh after BPTB ACLR. PROM cut-off values were met, but further rehabe is recommended to reduce functional deficits in BIA test and decrease risk of re-injury
Schilling, D et al. [27] (2020)	21	collegiate athletes	33.3	ACLR	THD, YBT, side plank, double leg lower test, SL squat, SL landing task from a height	Decrease in vertical ground reaction force on operative limb, less knee flexion and more valgus during drop landing task compared to non-operative limb, significantly reduced YBT score and reduced THD in cleared participants
Simon, J et al. [119] (2020)	236	15.7 ± 1.4	–	Lateral ankle sprain 22 (50.0) Medial ankle sprain 2 (4.5) Tibialis anterior strain 2 (4.5) Medial collateral ligament knee sprain 11 (25) Hamstring thigh strain 5 (11.5) Cartilaginous knee injury 2 (4.5)	SLHD	Limb symmetry did not significantly change pre- to post-injury with a mean difference of 1.5%. Following a lower extremity injury, single-leg hop for distance performance degrades not only for the injured limb but also the uninjured limb. However, limb symmetry did not change following a lower extremity injury.
Strong A, et al. [120] (2024)	24	ACLR-24.2(4.7), CTRL-22.4(3.1)	11 m 13f	ACLR	PROMs(TSK, IKDC, KOOS, Lysholm, Tegner), squat (bilateral bodyweight)	beyond the early phase of rehab from ACLR, individual level analyses are required to reveal differing loading strategies during bilateral squates
Sueyoshi, T et al. [28] (2017)	29	HS/college	41.3	ACLR	SLHD, THD, CHD, timed 6-m hop, *iso*-K Q/H strength	SL hop tests and *iso*-K were both useful for bilateral comparison of knee function; knee flexion deficits and flexion to extension ratios correlated with SL Hop performance
Suits W.H., et al. [121] (2024)	93			ACLR	single leg hop, triple hop, single leg vertical jump, lower extremity functional test	decline in performance measures in patients following ACLR who did not attend regular in person PT sessions in late stage rehab due to COVID-19 restrictions.

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Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Swindell H, et al. [122] (2023)	15	27.9 ± 7.6	6 m 9f	FAIS	dynamic ROM (3 plans for hip, knee, ankle and pelvic segments), single leg squat, PROs (HOS-ADL, HOS-Sports, IHOT-12, VAS)	Dynamic and saggital plane hip and knee ROMs improve after hip arhtroscopy for FAIS. These demonstrate strong correlations with improvements in HOS-ADL and 1 yr HOS-Sports scores
Tavares, M.L. A., et al. [123] (2023)	131	29.9 ± 7.7	115/16	ACLR	ACL-QoL, IKDC, GRS, ACL-RSI, TSK-17, isokinetic dynamometer, SLHD, THD, CHD, 6 m timed, biodex balance system	ACL-QoL showed moderate correlation with IKDC, GRS, ACL-RSI, and TSK-17. ACL-QoL had none to low correlations with muscle strength, postural stability, and hop tests. Knee functional status and psychological factors were found to be predictors of QoL
Teichmann J, et al. [124] (2022)	15	23 (6.0)	mostly male	unilateral knee injury	unilateral DJ test, unilateral isometric leg press test	differences in degree of asymmetry between control and rehab groups for contact time, flight time, and reactive strength index. Reactive strength data and use of dynamic assessment tools inform RTS decisions to limit potential for reinjury
Thompson X. D., et al. [125] (2022)	299	146/153	23 ± 9.7	ACLR	isokinetic dynamometer, SLDH, THD, (LSI pass = >90%), PROMs (IKDC, Tegner)	ACLR patients with higher activity levels may be more likely to pass hop tests despite failing quadri.tif strength test. May be an indicator of movement compensations to achieve jump symmetry in the presence of quadri.tif weakness
Toale, J et al. [126] (2021)	RTS: 1140 No RTS: 222	RTS: 23.6 ± 7.0 No RTS: 27.2 ± 7.5	–	ACLR	ACL-RSI, IKDC	The majority of patients that report they have not returned to play do so due to external life and psychological factors associated with their injury, including fear of reinjury and lack of confidence in performance. A small minority of patients were unable to return due to residual knee symptoms or reinjury
Ueda Y, et al. [127] (2022)	144	25.8 ± 11.9	82/52	ACLR	isokinetic quad strength (dyna), isokinetic hamstring strength (dyna), single leg hop distance, PROMs (IKDC, ACL-RSI)	LSI > 90%, IKDC score > 90. Meeting RTS criteria was positively associated with psychological readiness and patients who met multiple criteria had higher psychological readiness.
VanZile A, et al. [128] (2022)	34	16.5 ± 0.5(14–19)	18/17	ACLR with and without meniscal involvement	Y balance, single leg hop, triple hop	deficits in dynamic balance and hop performance among adolescent athletes who have undergone ACLR are not dependent on meniscal injury/surgery history
VanZile A, et al. (2022)	34	16.5 ± 0.5(14–19)	18/17	ACLR with and without meniscal involvement	Y balance, single leg hop, triple hop	deficits in dynamic balance and hop performance among adolescent athletes who have undergone ACLR are not dependent on meniscal injury/surgery history
Weber M, et al. [130] (2024)	34	24.2 ± 8.2	18/16	ACLR	vertical hop, single leg hop for distance, side hop, "Quality First" assessment	Quality of movement indicated no correlation with LSI from jump tests, therefore movement quality should also be evaluated in RTS tests.

Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Webster, K et al. [131] (2020)	450	24.0 ± 7.0	60.9	ACLR	ACL-RSI (≥65), SLHD, CHD and Iso-K Q strength. IKDC (>85). LSI (≥90%),	At 6 months after ACLR, few patients met al. l of the thresholds of the common tests used to assess RTS ability, although younger patients had higher rates of passing the functional tests. Self-perceived symptoms/function and psychological readiness were associated with a return at 12 months
Welling, W et al. [30] (2018)	62	24.2 + 6.2	72.6	ACLR	LESS, SLHD, THD, side hop, *iso*-K Q/H strength, IKDC, and ACL-RSI	Low percentage of passing all RTS was low at both 6 and 9 months after ACLR, quad strength deficits persistent during this timeframe, IKDC questionnaires shows insufficient self-reported knee function for RTS
Welling, W et al. [132] (2020)	64	27.8 ± 8.8	68.7	ACLR	LESS, SLHD, THD, side hop test, *iso*-K Q/H	Better jump landing patterns, hop performance and greater hamstring strength greater likelihood for RTS, but these criteria failed to identify patients are at risk for 2nd ACL injury
Werner, J et al. [37] (2018)	RTS: 28 No RTS: 12	RTS: 23 ± 11 NRTS: 26 ± 13	RTS: 60.7 NRTS: 41.7	ACLR	PRO (IKDC, KOOS, Tegner Activity Scale, MARS) Star excursion balance (Ant. and PM), SLHD, CHD, 6-m timed hop, *iso*-K Q/H strength, step down to fatigue test	Non-RTS group demonstrated lower PRO despite demonstrating similar functional performance; self-perceived function also importance factor for RTS
White, A et al. [31] (2018)	20	23.7 ± 3.0	20.0	healthy	Two-minute lateral step-down test vs 30-second side-hop test prior to SLHD	Two-minute lateral step-down test resulted in a greater decrease in hop performance compared to the 30-seconds side-hop test. Two-minute lateral step-down test may be an effective method of inducing fatigue to better mimic performance in a sports environment to inform RTS decisions
Worner, T et al. [48] (2019)	Hip Arthroscopy (HA): 33 Control: 33	HA: 32.3 ± 9.4 Control: 31.1 ± 10.6	HA: 87.8 Control: 87.8	Hip arthroscopic surgery	HAGOS, hip range of motion (active and passive flexion, passive internal rotation, passive external rotation), *iso*-M abduction, adduction, flexion, extension, internal rotation, and external rotation strength, YBT, medial and lateral triple hop test for distance, Illinois agility test	Subjective hip function was substantially impaired in patients 6 to 10 months after hip arthroscopy in comparison to healthy controls. The hip arthroscopy group presented with comparable objective hip function for the majority of outcomes except of hip range of motion and hip flexion strength. No consistent pattern of impairments was found in operated hips compared with non-operated hips.
Xergia, S et al. [133] (2014)	22	28.8 ± 11.2	100	ACLR	*iso*-K Q/H strength, SLHD	SLHD has potential to identify muscle strength asymmetries but not kinetic or kinematic asymmetries 6–9 months post ACLR

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Table 2 (*continued*)

Author and year	# of Subjects	Age	Male Sex in %	Diagnosis	Tests	Conclusion/Criteria for return to sport
Zarro M, et al. [134] (2023)	21	20.38 ± 1.67	7 m 14f	ACLR	ACL-RSI, SHD, THD, CHD, 6-m timed, SLVH,	SLVD, SHD, THD, and CHD are correlated with ACL-RSI and improvements in physical function during rehab may improve psychological status and optimize RTS after ACLR
Zhou L, et al. [135] (2022)	30	27.7 ± 5.9	26/4	ACLR	PROMs(IKDC, ACL-RSI), isokinetic dynamometer, single and triple hop, athrometer for knee laxity, 6DOF kinematics	IKDC within 15th %ile for age and sex matched subjects. ACL-RSI > 56. LSI > 90% peak torque for Q and H. LSI > 90% for SH and TH. Bilateral displacement difference < 3 mm. Difference in ROM between limbs < 10% in 6DOF.
Zwolski, C et al. [136] (2016)	45	10–30	–	ACLR	SLHD, THD, CHD, *iso*-M Q strength	LSI not appropriate for bilateral ALCRs

3.3. Knee

Eighty five percent (97/114) of articles discussed RTS for knee pathologies. A list of common functional tests analyzed in the studies can be found in Figure 2. Furthermore, 94 out of the 114 (82%) specifically analyzed RTS following anterior cruciate ligament reconstruction (ACLR). These studies looked at both ipsilateral injuries as well as contralateral injuries as 50% of 2nd ACL injuries occur in the contralateral limb [14,15].

3.3.1. Strength testing

Isokinetic dynamometry was utilized in 44/114 (39%) articles analyzing the knee in this review. Numerous individual functional tests were utilized among the articles in this review. Isokinetic dynamometers are passive devices which resist applied forces, control the speed of exercise at a predetermined rate, and provide a record of applied force throughout a joint range of motion [16]. Suggested benchmarks for RTS values with isokinetic testing are 85% Limb Symmetry Index (LSI) for quadriceps, 90% LSI for hamstrings, and Hamstring:Quadricep (H:Q) ratio of greater than 70% [17]. Gokeler et al. suggests LSI greater than 90% on both quadriceps and hamstrings as well as a H:Q ratio greater than 55% for females and 62.5% for males [18]. There have been several studies validating isometric dynamometry with handheld dynamometry (HHD) against isokinetic [19–22] as this is a more readily available option for outpatient clinics to utilize. As described above, isokinetic dynamometers are computerized machines capable of providing multiple elements of measuring muscle strength, including peak force, endurance, power, angle of maximal force, and are capable of generating strength curves; while, HDD is a convenient device that can be placed between the hand of the practitioner and the patient's tested body part, similar to how a practitioner would perform a manual muscle test [23]. In a 2011 systematic review, it was determined the HDD's ease of use, portability, cost, and compact size, compared with traditional isokinetic devices can be regarded as a comparable, reliable and valid instrument for muscle strength assessment in a clinical setting [23]. Furthermore, the current review shows results

Knee Functional Test
1) **Strength with Hand-Held Dynamometry (or Isokinetic is available)** • **Quadriceps** • **Hamstrings**
2) **Y-Balance**
3) **Single Leg Vertical**
4) **Single Leg Hop with Fatigue Protocol**
5) **Cross-Over Hop**
6) **Figure 8 Hop**
7) **Drop Landing Knee Excursion**
8) **Single Leg Press**

Figure 2. Shows a suggested test battery that can be utilized to assess patients with knee pathologies.

that supports the hypothesis: the HHD provides a valid measure of quadriceps peak torque and rate of torque development compared to traditional isokinetic dynamometers [22].

3.3.2. Hop tests

Historically, hop tests provide a reliable and valid performance-based outcome measure for patients undergoing rehabilitation following ACL reconstruction [11]. With this said, there have been multiple versions of the hop test that have been studied and described hereafter. In our search, Single Leg Hop for Distance (SLHD) was mentioned in 56 (58%) articles for the knee. Overall, it seems to be agreed upon that although it is not sensitive enough to be used as sole criterion for RTS, it does have strong correlation with knee extensor strength. Though a specific SLHD passing criteria lacks, Kotsifaki et al. discovered that during landing, the reconstructed limb had less peak knee flexion, internal rotation and knee flexion ROM (4.2°, 3.4° and 3.9°, respectively) when compared with the contralateral leg, with larger effect size (ES) and absolute values for peak knee internal rotation and knee flexion ROM when compared with controls [24]. Furthermore, Birchmeier et al. reported in addition to SLHD indicating improved knee extension strength, the Triple Hop for Distance (THD) has indications of improved reactive strength [25]. THD was discussed in 33 (34%) articles. THD has been determined to be a strong predictor of lower limb muscular strength and power; however, it cannot predict static balance ability [26]. Another study with collegiate athletes who had undergone ALCR and have been cleared to RTS, found there is reduced THD LSI even in the cleared athletes [27]. Sueyoshi et al. in 2017 compared all four of the SL hop tests to isokinetic strength testing and stated that knee flexion and flexion to extension ratios were correlated with SL hop performance [28].

Next, the single leg vertical jump (SLVJ) was seen in 8.2% (8/97) of the articles and may also be a beneficial test to use when evaluating RTS for ACLR athletes. There have been significant correlations between SLVJ and subjective International Knee Documentation Committee Questionnaire (IKDC), Tegner Activity Scale, Anterior Cruciate Ligament-Return to Sport After Injury Scale (ACL-RSI), and isokinetic extensor muscle strength as well as other functional tests (SLHD, co-contraction, shuttle run, and carioca tests.) [29]. Lastly, the side hop test (SH) was utilized in 14 (14%) articles, primarily as a part of functional test batteries. Gokeler et al. and Welling et al. in 2017 found a majority of athletes are not ready to RTS at 6-month post-operative using a test battery consisting of the Landing Error Scoring System (LESS,) SLHD, THD, SH, and isokinetic testing [18]. Subsequently, Welling et al. in 2018 reported only 11.3% of ACLR patients will pass all of the same test battery including the side hop test at 9 months [30]. This demonstrates there is still a measurable difference that can be clinically discerned with the functional tests utilized. Using a two minute fatigue protocol prior to SLHD as described by White et al. resulted in a greater decrease in hop performance [31]. The authors concluded that this two minute lateral step test may be an effective method of inducing fatigue to better mimic performance in as sports environment to inform RTS decision making. Altogether, a myriad of hop tests are available to examine patient's ability to RTS; however, the ultimate passing criteria to minimize re-injury and expedite RTS has yet to be determined.

3.3.3. Drop landing

The drop landing test was analyzed in 5 (5.15%) articles. In 2015, Ithurburn et al. described the drop landing test as a single limb drop off a 31 cm box and landing on the same limb with a controlled landing for at least 3 s. Using motion capture analysis, they found at the time of RTS, athletes who had undergone ACLR demonstrated asymmetry in landing mechanics compared to healthy controls. Compensations included increased trunk flexion, decreased knee flexion excursion, and decreased knee extension moments [32]. Schilling and Radwan in 2020 also found less knee flexion and more valgus during drop landing task compared to non-operative limb in collegiate athletes after an ACLR [27]. Wilczyński et al. revealed decreased knee flexion excursion and dynamic knee valgus as important risk factors to monitor following ACLR to prevent re-injury [33]. Ultimately, this test seems to capture both well.

3.3.4. Balance testing

To begin, balance testing was used in 17 (18%) articles. It has been found that single leg balance in straight knee position is not predictive enough to detect impairments in post-ACLR population and is not predictive of subjective knee function [34]. Garrison et al. in 2015 reported the dynamic assessment of balance and quad strength with the Y-Balance Test (YBT) in the anterior direction to be predictive of LSI at time of RTS. In our review, the YBT was seen in 12% (12/97) of articles. When looking at YBT, a deficit of greater than 4 cm at 12 weeks post-op was predictive of athletes not achieving 90% LSI at for SLHD and THD at time of RTS [35]. Other authors have shown significant reduction in YTB at RTS [27,36]. The Star Excursion Balance Test was also used by Werner in 2018, however, they found no significant difference in functional performance in the RTS group and group who was unable to RTS [37]. Y-Balance test may be a good test of functional lower extremity strength and balance to use at time of RTS and also beneficial to use at 12 weeks to assess progress and predict likelihood of 90% LSI with hop testing at the 6-month mark [35] (Table 3).

3.3.5. Patient reported outcomes

Increasing emphasis has been placed on patient reported outcomes (PROs) and overall psychological readiness to return to sport in recent studies for ACLR population as seen by its high incidence in reviewed articles. See Table 4 for a comprehensive list of PROs utilized among the articles in this review. Albano et al. found psychological readiness assessed with both ACL-RSI and IKDC to be the most important factor in subjects who did not RTS followed by agonist: antagonist (hamstring to quadriceps) ratio [38]. Generally higher knee strength and hop test performance is associated with increased self-reported

Table 3
Incidence of Functional Tests in Reviewed Articles.

Test	Incidence in Review (%)
Single Leg Hop for Distance (SLHD)	48
Isokinetic Strength Testing (*iso*-K)	38
Triple Hop for Distance (THD)	28
Isometric Strength Testing (*iso*-M)	28
Cross-over Hop for Distance (CHD)	20
6-meter Timed Hop	15
Side Hop Test (SH)	12
Y-Balance Test (YTB)	10
Single Leg Drop Landing	6
Single Leg Step Down	5
Landing Error Scoring System (LESS)	5
Medial Triple Hop For Distance	5
Lateral Triple Hop For Distance	4
Single Leg Squat	3
Vertical Jump	3
Single Leg Vertical Jump (SLVJ)	3
KT-2000 Arthrometer	3
Drop Vertical Jump	3
KT-1000 Arthrometer	3
Countermovement Jump	3
Timed Square Hop	3
Biodex Multi-joint system – 4	3
Co-contraction Test	2
Carioca Test	2
Shuttle Run Test	2
Single Leg Balance	2
Back in Action Test Battery (BIA)	2
Vail Sport Test	2
Leg Press	2
Isotonic Strength Testing (*iso*-T)	2
Patient-Acceptable Symptom State (PASS)	2
Bilateral Jump Landing	2
Step Down to Fatigue	2
Star Excursion Balance Test (SEBT)	2
Balance System SD Dynamometer	2
Running Speed	2
Trunk Endurance (Plank)	2
Timed Figure 8 Hop	2
Timed 6 Meter Crossover Hop	2
6 Meter Crossover Hop For Time	1
Single Leg drop vertical	1
Qualitative Assessment of Single-leg Landing (QASLS)	1
Modified Illinois Change of Direction Test (MICODT)	1
Crossover Reach Test For Distance	1
Stanading Heel Raises	1
Stork Test (With & Without BOSU)	1
Quadrant Hop (Clockwise & Counterclockwise)	1
Single Leg Drop Land & Side Cut	1
Single Leg Drop Land & Forward Run	1
20 Meter Single Leg Timed Hop	1
20 Meter Single Leg Timed Crossover Hop	1
Double Leg Lower Test	1
Side Plank	1
2 Minute Lateral Step Down Test	1
6DOF Kinematics	1
"Quality First" assessment	1
Unilateral Drop Jump	1
Stretch-Shortening Cycle	1
Agility T Test	1

outcomes [39–42]. However, RTS groups vs non-RTS groups have similar functional performance, but there is a significant difference only in PRO suggesting self-perceived function is an important for RTS [37].

3.4. Hip

The current review included seven articles that studied RTS guidelines specific to the hip. Rath et al. performed an international cross-sectional survey of 26 high-volume hip arthroscopy specialized surgeons to report their weight-bearing lim-

Table 4
Incidence of PROs in Reviewed Articles.

Scale (Knee)	Incidence in Review (%)
International Knee Documentation Committee Questionnaire (IKDC)	38
Anterior Cruciate Ligament-Return to Sport After Injury (ACL-RSI)	28
Knee Injury and Osteoarthritis Outcome Score (KOOS)	16
Tegner Activity Scale	12
Lysholm Score	10
Knee Outcome Survey-Activities of Daily Living Scale (KOS-ADLS)	2
Knee Outcome Survey-Sports Activity Scale	2
Knee Injury and Osteoarthritis Outcome Score-Quality of Life Subscale (KOOS-QoL)	2
Scale (Hip)	**Incidence in Review (%)**
Hip Outcome Score-Sport Scale (HOS-SS)	3
Modified Harris Hip Score (MHH)	2
Nonarthritic Hip Score	2
Copenhagen Hip and Groin Outcome Score (HAGOS)	2
International Hip Outcome Tool (IHOT-12)	2
Scale (other)	**Incidence in Review (%)**
Tampa Scale of Kinesiophobia form (TSK-11)	7
Marx Activity Rating Scale (MARS)	3
Global Rating Scale of Perceived Functional (GRS)	2
Pain Catastrophizing Scale (PCS)	2
Visual Analog Scale (PAS)	2

itations and rehabilitation protocols after various arthroscopic hip procedures. They determined most surgeons allowed patients to return to running between 3 and 4 months post-operatively and almost 83% of surgeons agreed microfracture procedures required further delay to return to running and impact sports [43]. The results captured in this study were limited to timeline from surgery date. In 2014, Domb et al. surveyed twenty-seven orthopedic surgeons from high-volume hip arthroscopy centers regarding RTS after hip arthroscopy. They were asked to give a time frame for RTS and choose meaningful criteria. They found that regarding time frame for RTS, 70% of surgeons recommended between 12 and 20 weeks. In addressing criteria, 85% of surgeons recommended patients need to be able to produce all motions involved in their respective sport without pain and a majority of surgeons agreed pain-free running, jumping, lateral agility drills, and single-leg squats should be included [44]. Monahan et al. in 2021 reported athletes with a postoperative alpha angle $\leq 46°$ had a 6.3 times greater chance of returning to sport at than those with a postoperative angle $>46°$ [45].

These authors agree the criteria listed above are all important but argue there is a need for greater objectivity when testing these athletes. There is good evidence in the medial triple hop test which was found to be significantly reduced in the FAI population, and therefore, may be beneficial in identifying side to side asymmetries at time of RTS [46,47]. In 2013, Kivlan et al. studied dancers with unilateral hip pain utilizing the cross-over reach, medial triple hop, lateral triple hop, and cross-over hop tests. They reported performance of the medial triple hop was the only test that demonstrated a significant difference on the involved side compared to the uninvolved side [47]. In 2016, Kivlan et al. again revealed that the medial triple hop was significantly reduced in FAI group compared to healthy controls and discovered similar results with the lateral triple hop was reduced in this population as well. Similarly, hip extension strength deficit measured with hand-held dynamomter (HHD) was found in dancers with FAI [46]. Worner et al. in 2019 studied function after hip athroscopy with objective criteria including: range of motion (ROM; flexion, internal rotation, external rotation), isometric hip strength (adduction, abduction, flexion, internal rotation, and external rotation), and performance-based measures (Y-Balance Test, medial and lateral triple-hop test, and Illinois agility test. Objective measures of hip ROM, hip flexion strength, and posteromedial reach of the Y-Balance Test were reduced in the hip arthroscopy group, although not statistically significant. The only significant difference between groups was reduced passive hip flexion [48]. Hip external rotation strength as measured with HHD by Malloy et el. was correlated with better dynamic control of the lower extremity during unanticipated single leg landing and cutting tasks [49]. Kivlan et al. in 2012 also reports diminished deep squat depth and provocation or pain during the single-leg balance test in patients with FAI and also convergent validity in the star-excursion balance test (SEBT) and single-leg squat tests through kinematic analysis [50].

Although it was not the aim of this review to address psychological readiness, it has been found to be an important component in RTS. Worner et al. stated self-reported hip function using Copenhagen Hip and Groin Outcome Score (HAGOS) was reduced compared with controls at 6–10 months after surgery suggesting there may be beneficial to utilize PROs prior to RTS [48]. Monahan et al. also described athletes who returned to sport after hip arthroscopy for FAI achieved the minimal clinically important difference for the Hip Outcome Score-Sport Specific Subscale at significantly higher rates than athletes who did not RTS [45]. Browning et al. in 2021 using the Tampa Scale of Kinesiophobia (TSK-11), Pain Catastrophizing Scale (PCS), and Hip Outcome Score-Sport Specific Scale (HOS-SS) found that patient kinesiophobia and pain catastrophizing at 1-year

follow-up were negatively correlated with RTS and achievement of a clinically significant outcome in sport-related activities at 2-year follow up [51].

3.5. Ankle

From the studies identified by the current review which incorporated ankle injuries, only 3 studies met the inclusion criteria by evaluating functional tests. Lunsford et al. concluded the single leg heel-rise test to be a valuable assessment tool for gastrocnemius and soleus strength [52]. They found the average number of heel-rises was 27.9 (SD = 11.1, minimum = 6, maximum = 70) for all groups, both genders, no differences between male and female subjects, and with the lower 99% confidence interval at 25. Their recommendation is to use 25 repetitions as the normative value for plantar-flexion strength using the standing heel-raise test [52]. Caffrey et al. looked at four single limb hopping tests; Figure 8 hop test, side hop test, 6-meter crossover hop test, and square hop test related to functional ankle instability (FAI) [53]. There are significantly measurable functional performance deficits present in each of the four tests on the FAI limb compared to the contralateral uninjured limb. Clanton et al. suggested return to sport testing should include balance and proprioception, strength, range of motion and agility as well as psychological assessment of the athletes' readiness to return to sport [54]. Specifically they advocate for the dorsiflexion lunge test as a measure of functional range of motion which has confirmed intra- and interrater reliability and in some research has been predictive of future injury in cricket and soccer. For balance and proprioception testing, the authors discuss the star excursion balance test (SEBT) or the instrumented version with three directions of movement instead of eight; the Y Balance Test (YTB). The SEBT was shown to have excellent intra-observer reliability and sensitive for FAI. The SEBT was also a predictive measure of lower extremity injury in high school basketball players. High reliability was demonstrated for the Agility T-Test and typical times were found between 8.9 and 13.5 s. Finally, the Sargent/Vertical Jump test evaluated functional strength, power, and dexterity and was reported to be a reliable test for estimation of explosive power in the lower limb.

4. Discussion

The knee, more specifically ACL injuries, have been the most researched for RTS testing. The results highlight the importance of performing a battery of tests focused on multiple facets of strength, neuromuscular control, agility, power, and landing biomechanics before considering medical clearance and RTS. Majority of authors agree that quadriceps strength in particular is of vital importance. Because the gold standard (*iso*-kinetic testing) not being available in most clinics, several studies validate the use of handheld isometric dynamometry against isokinetic. Another alternative includes isotonic testing due to its availability and capacity to act as a functional method of testing as seen with a leg press test or step-down test. Secondly, hop tests are considered a widely necessary functional movement capability test that needs to be assessed prior to RTS. The SLHD demonstrates low sensitivity when used in isolation while attempting to identify limb asymmetry prior to RTS; however, it does have a strong correlation with knee extensor strength. Evidence suggests that additional hop tests, the THD, SLVJ, and side hop test, may all be beneficial at determining RTS readiness. When utilizing these tests for RTS consideration, form should be evaluated. Specifically for the drop landing test, motion capture technology will assist in identifying asymmetries and compensations, such as increased trunk flexion, decreased knee flexion excursion, and decreased knee extension moments, which are all risk factors for ACL injuries. Third, balance testing allows for proprioception assessment which can be markedly affected after injury and needs to be properly restored prior to RTS. One such test, the Y-Balance Test, is a good predictor for functional lower extremity strength, balance, and predictor of RTS. Finally, patient reported outcomes is an enormous factor for successful RTS and should be measured at time of testing.

Aside from the knee, there is a paucity of research for RTS testing for the hip and ankle, but there is good evidence for a few specific metrics. Hip RTS protocols should consider the medial triple hop test, extension and external rotation strength testing with HHD, decreased single leg squat depth or pain provocation during single leg balance tests, and decreased passive hip flexion. These were all found to be significantly diminished or painful with the FAI population and should be evaluated prior to RTS. For the ankle, a number of specific, beneficial tests to evaluate the strength and functional stability of the ankle in multiple planes of motion were identified. These are the Figure 8 hop test, side hop test, 6-meter crossover hop, and square hop test. Measurable differences were identified between healthy participants and participants with known functional ankle instability. There is also evidence to suggest the dorsiflexion lunge test is a good functional measure of ankle mobility and should be no less than 9–10 cm; the SEBT or Y Balance test are reliable measures of functional ankle balance/proprioception; the Agility T-Test has high reliability for agility; and the Sargent/Vertical Jump test is reliable for the estimation of explosive power in the lower extremity. As with the knee, psychological patient reported outcomes should be unitized for both the hip and ankle as well.

Additionally, we have included suggested test batteries for the knee, hip, and ankle based on current literature and our practice as demonstrated in Figs. 2–4. Note, this is an ever-changing science and more research validating individual tests needs to be performed.

Hip Functional Test
1) Strength with Hand-Held Dynamometry • Hip Abduction • Hip Extension • Hip External Rotation
2) Y-Balance
3) Single Leg Vertical
4) Single Leg Hop with Fatigue Protocol
5) Cross-Over Hop
6) Medial Triple Hop
7) Figure 8 Hop
8) Single Leg Press

Figure 3. Shows a suggested test battery that can be utilized to assess patients with hip pathologies.

Ankle Functional Test
1) Single Leg Heel Raise Test
2) Y-Balance
3) Single Leg Vertical
4) Single Leg Lateral Hop with Fatigue Protocol
5) Cross-Over Hop
6) Figure 8 Hop
7) Side Hop Test
8) Square Hop Test
9) Single Leg Press

Figure 4. Shows a suggested test battery that can be utilized to assess patients with ankle pathologies.

5. Limitations

This review serves as a guide to current best evidence-based practice where RTS testing is concerned, but certainly demonstrates the need for further investigation on RTS testing validity. Specifically, all the tests in this search used limb symmetry index or "validated" the test against another metric (ie. Hop test related to strength,) but none of them followed athletes once given RTS clearance to assess re-injury rates. Additionally, it was discovered the majority of studies analyzing return to sport testing were under increased risk for bias due to researchers not controlling for type of surgical procedure and/or graft, lack of control for concomitant injury, not incorporating both sexes and/or analyzing the differences between the sexes, and/or not considering level of activity and/or sport the patient participated in before and after the study. With this said, it is important for patients who have successfully returned to sport after injury to be followed for the remainder of their competitive athletic careers for the opportunity to correlate which tests were successful in reducing the risk of future re-injury, and which tests completed at time of RTS were not predictive of injury recurrence.

6. Conclusion

Even with the enormous amount of literature that exists regarding ACL injuries and testing, there is no standardized criterion for RTS clearance. Alternatively, the research available for the hip and ankle RTS is severely lacking. Therefore, the authors suggested lower extremity joint-specific test batteries from this review that can serve as a framework for future research and validation for joint-specific RTS functional testing.

Ethics approval

Ethical approval was not sought for the present study because data from previously published studies in which informed consent and ethical approval was obtained by each study's primary investigators will be utilized for the present study's analysis.

CRediT authorship contribution statement

Traci Smiley: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Johnathan Dallman:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation. **Rachel Long:** Writing – review & editing. **Mason Kapple:** Writing – review & editing. **Levi Aldag:** Writing – review & editing, Methodology, Investigation. **Anthony Mok:** Writing – original draft, Methodology, Investigation. **Christopher Bernard:** Writing – review & editing, Supervision, Project administration, Investigation. **Kyle Martin:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Lisa Vopat:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Bryan Vopat:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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